BRITISH STANDARD • BSI Standards Publication

BS EN 18246:2026

中英双语对照版

Digital product passport — Data authentication, reliability and integrity

数字产品护照 — 数据认证、可靠性与完整性

ICS 35.240.63 | ISBN 978 0 539 37850 4 | 2026-09-30 发布
CEN 于 2026-08-17 批准 | 最迟 2027-03 转为成员国国家标准
编制机构：CEN-CENELEC/JTC 24（秘书处 DIN）

内部学习用 | 中英对照 | 上英下中

© 2026 CEN/CENELEC & BSI。本对照版仅供内部学习，正式合规以 CEN/BSI 英文原版为准。

目录 / Contents

英国前言 / National foreword 03

欧洲前言 / European foreword 03

引言 / Introduction（0.1 背景 / 0.2 概述） 04

1 范围 / Scope 05

2 规范性引用文件 / Normative references 05

3 术语和定义 / Terms and definitions（3.1–3.16） 06

4 DPP 通用安全假设 / General security assumptions for DPP 08

4.1 通则 | 4.2 系统参与方 | 4.3 公开数据访问 | 4.4 通信 08

4.5 载体内标识 | 4.6 载体真实性 | 4.7 DPP 真实性与数据完整性 09

5 唯一标识、载体与 DPP 数据的风险与要求 10

5.1 数据安全与隐私（5.1.1–5.1.8） 10

5.2 标识风险与要求（5.2.1–5.2.2） 12

5.3 产品风险（5.3.1–5.3.2） | 5.4 防排斥（5.4.1–5.4.3） 12

Annex A（规范性）电子签名数据构造（ESDC） 13

Annex B（资料性）ESDC 示例 14

B.2 属性电子证明 | B.3 可见数字印章 VDS | B.4 AIDC DigSig 14

B.5 W3C 可验证凭证 | B.6 数字签名与电子印章 15

Annex C（资料性）公平竞争相关风险与考量 16

Annex ZA（资料性）与法规 (EU) 2024/1781 的关系 16

参考文献 / Bibliography（[1]–[23]） 17

英国前言 / National foreword

This British Standard is the UK implementation of EN 18246:2026. The UK participation in its preparation was entrusted to Technical Committee IST/47/1. This publication is provided as is, and is to be used at the recipient's own risk. BSI makes no representation or warranty as to adequacy, accuracy, completeness or reasonableness. This publication is not intended to constitute a contract. Users are responsible for its correct application.

> 本英国标准系 EN 18246:2026 的英国实施版本。英国参与编制的工作委托给技术委员会 IST/47/1。本出版物按"现状"提供，由使用者自行承担使用风险。BSI 不对充分性、准确性、完整性或合理性作任何明示或暗示的陈述与保证。本出版物不构成合同，使用者对其正确应用负责。

This publication has been prepared under a mandate given to the European Standards Organizations by the European Commission and the European Free Trade Association. It is intended to support requirements of the EU legislation detailed in the European Foreword. A European Annex (usually ZA or ZZ) describes how this publication relates to that EU legislation. For the Great Britain market, if UK Government has designated this publication for conformity with UKCA marking legislation, it may contain an additional National Annex showing correlation with UK legislation. For Northern Ireland, EU law continues to apply; Annex ZA/ZZ remains valid.

> 本出版物依欧盟委员会与欧洲自由贸易联盟授予欧洲标准组织的授权编制，用于支撑欧洲前言中所述欧盟立法要求。欧洲附录（通常为 ZA 或 ZZ）说明本文件与该欧盟立法的关系。大不列颠市场若由英国政府指定用于 UKCA 标识立法，可附加国家附录标示与英国法的对应关系；若无此类国家附录，欧洲正文中的 ZA/ZZ 附录标示与适用于大不列颠的英国法规的关系。北爱尔兰市场继续适用欧盟法，ZA/ZZ 附录及欧盟法引用仍然有效。

Metadata：ICS 35.240.63 ｜ ISBN 978 0 539 37850 4 ｜ 2026-09-30 在标准政策与战略委员会授权下发布 ｜ 符合英国标准不免除法律义务。

欧洲前言 / European foreword

This document (EN 18246:2026) has been prepared by the Joint Technical Committee CEN-CENELEC/JTC 24 "Digital Product Passport – Framework and System", the secretariat of which is held by DIN. This European Standard shall be given the status of a national standard, either by publication of an identical text or by endorsement, at the latest by March 2027, and conflicting national standards shall be withdrawn at the latest by March 2027.

> 本文件（EN 18246:2026）由 CEN-CENELEC 联合技术委员会 JTC 24"数字产品护照—框架与系统"编制，秘书处由 DIN 承担。本欧洲标准最迟应于 2027 年 3 月前，以发布等同文本或认可的方式获得国家标准地位，冲突的成员国国家标准最迟于 2027 年 3 月撤销。

Attention is drawn to the possibility that some elements may be the subject of patent rights; CEN-CENELEC shall not be held responsible for identifying any or all such patent rights. This document has been prepared under a standardization request addressed to CEN-CENELEC by the European Commission; the Standing Committee of the EFTA States subsequently approves these requests for its Member States. For relationship with EU Legislation, see informative Annex ZA, which is an integral part of this document. Feedback should be directed to users' national standards bodies.

> 请注意本文件某些内容可能涉专利权利，CEN-CENELEC 不承担识别全部专利的责任。本文件依欧盟委员会向 CEN-CENELEC 提出的标准化请求编制；欧洲自由贸易联盟常设委员会随后为其成员国批准这些请求。与欧盟立法的关系见资料性附录 ZA，其为本文件的组成部分。反馈请提交至各成员国国家标准机构。

本欧洲标准有三种官方版本（英、法、德）。由 CEN/CENELEC 成员自行翻译为其他语言并通知管理中心的版本，与官方版本具有同等地位。CEN-CENELEC 管理中心：Rue de la Science 23, B-1040 Brussels。

引言 / Introduction

0.1 背景 / Background

A digital product passport (DPP) is a key enabling mechanism to make product information traceable and accessible across value chains – supporting economic operators, manufacturers, distributors, repairers, recyclers and consumers to make informed decisions and to support a circular economy. The implementation of DPPs will be carried out progressively. Sector-specific initiatives will determine the precise DPP content and requirements for individual product groups.

> 数字产品护照（DPP）是使产品信息在价值链中可追溯、可获取的关键赋能机制，支撑经济运营商、制造商、分销商、维修商、回收商与消费者做出知情决策，并支撑循环经济。DPP 将逐步实施，具体产品组的 DPP 内容与要求由分行业倡议确定。

To support the implementation of DPPs, 8 standards have been developed:

> 为支撑 DPP 实施，共开发 8 项标准：

- EN 18219:2026 — Digital product passport – Unique identifiers | 唯一标识符
- EN 18220:2026 — Digital product passport – Data carriers | 数据载体
- EN 18216:2026 — Digital product passport – Data exchange protocols | 数据交换协议
- EN 18222:2026 — Digital product passport – APIs for product passport lifecycle management and searchability | 护照生命周期管理与可检索 API
- EN 18223:2026 — Digital product passport – System interoperability | 系统互操作
- EN 18221:2026 — Digital product passport – Data storage, archiving, and data persistence | 数据存储、归档与持久化
- EN 18239:2026 — Digital product passport – Access rights management, information system security, and business confidentiality | 访问权限、信息安全与商业保密
- EN 18246:2026 — Digital product passport – Data authentication, reliability and integrity（本文件）

0.2 概述 / Overview

Trust is key to any transaction whether physical or digital. Data and information collected during the whole life cycle of the product is to be protected from both accidental and malicious compromise and misuse.

> 信任是任何交易（无论是实体还是数字）的关键。产品全生命周期内收集的数据与信息应受到保护，防止意外与恶意的损害及误用。

Reasonable security expectation is likely to address the following elements of the DPP deployment: management and verification of identifiers; relationship between unique identifiers and authentication elements; identification of the verifier and authorized access to privileged product-related information; authentication solutions; artefact metrics (technical measurements supporting verification or trust in physical objects); integrity-preserving communication along the supply chain; and electronically signed data constructs (ESDCs) or equivalent functions.

> 合理的安全预期应覆盖以下 DPP 部署要素：标识管理与核验；唯一标识与认证要素之间的关系；验证方身份及授权访问产品特权信息；认证方案；实物度量（支撑实物对象验证或信任的技术测量）；沿供应链保护完整性的通信；以及电子签名数据构造（ESDC）或等效功能。

A wide variety of information security techniques are routinely deployed at scale. The EU DPP system should leverage industry standards and best practices in all aspects, from data generation and storage through to access rights management. The unique product identifier, as stored in a data carrier, can be

considered as the DPP access anchor. Trust in the identifier is therefore a foundational element to the trust of DPP.

> 各类信息安全技术已在大规模场景常规部署。欧盟 DPP 系统应在数据生成、存储到访问权限管理的各方面借鉴行业标准与最佳实践。存于数据载体中的唯一产品标识可视为 DPP 的"访问锚点"，因此对标识的信任是 DPP 信任的基础要素。

NOTE 2：ISO 22373、ISO 22380、ISO 22383 概述应对实物产品欺诈的原则，聚焦产品与文件的真实性、完整性与信任；强调理解欺诈情境（动机、行为）以选择实物认证方案。

1 范围 / Scope

This document defines the requirements and frameworks for secure information processing and communication to safeguard integrity, authenticity and reliability in the digital product passport (DPP) data exchange, minimizing product fraud and counterfeiting through data verification and integrity enforcement mechanisms.

> 本文件规定了安全信息处理与通信的要求和框架，以保障数字产品护照（DPP）数据交换中的完整性、真实性与可靠性，通过数据核验与完整性强制机制降低产品欺诈与假冒风险。

This document provides a framework for establishing trust, interoperability, and interoperation via secure electronically signed data construct (ESDC) for multi-actor applications, applicable across various sectors and in multilingual environments. Existing hardware and software systems for unique product identification and storage of this identification are to be considered.

> 本文件为多主体应用提供通过安全电子签名数据构造（ESDC）建立信任、互操作与协作的框架，适用于各行业与多语言环境。应兼顾用于唯一产品标识及其存储的既有软硬件系统。

Out of scope: system architecture for DPP; DPP use cases; secure elements related to data carriers; cryptographic security features for unique product identifiers.

> 不适用范围：DPP 系统架构；DPP 用例；数据载体安全元件；唯一产品标识符的密码学安全特征。

NOTE 1 While not disrupting existing traceability and authentication systems, this document facilitates interoperability by introducing an ESDC scheme to be combined with existing data constructs to cover and preserve existing data models.

> 注 1：本文件在不破坏既有追溯与认证系统的前提下，通过引入 ESDC 方案与既有数据构造组合，以覆盖并保留既有数据模型，从而促进互操作。

NOTE 2 Annex B includes illustrative examples and references to supporting implementations, intended to demonstrate approaches that promote interoperability across diverse environments.

> 注 2：附录 B 包含说明性示例及支撑实现参考，旨在展示促进跨不同环境互操作的方法。

---

2 规范性引用文件 / Normative references

The following documents are referred to in such a way that some or all of their content constitutes requirements. For dated references, only the edition cited applies. For undated references, the latest edition (including amendments) applies.

> 下列文件以某些或全部内容构成本文件要求的方式被引用。注日期引用文件，仅所引版本适用；未注日期引用文件，适用最新版（含修订）。

- EN 18216:2026 — Digital product passport – Data exchange protocols ｜ 数据交换协议
- EN 18219:2026 — Digital product passport – Unique identifiers ｜ 唯一标识符
- EN 18220:2026 — Digital product passport – Data carriers ｜ 数据载体
- EN 18221:2026 — Digital product passport – Data storage, archiving, and data persistence ｜ 数据存储、归档与持久化

- EN 18239:2026 — Digital product passport – Access rights management, information system security, and business confidentiality ｜ 访问权限、信息安全与商业保密

3 术语和定义 / Terms and definitions

For the purposes of this document, the following terms and definitions apply. ISO and IEC maintain terminology databases at https://www.iso.org/obp/ and https://www.electropedia.org/.

> 就本文件而言，适用下列术语与定义。ISO 与 IEC 分别在下列地址维护术语数据库。

3.1 electronically signed data construct (ESDC) 电子签名数据构造

structured data set containing the header, payload, signature and optional data fields

> 含头（header）、载荷（payload）、签名（signature）及可选数据字段的结构化数据集。

3.2 digital product passport (DPP) 数字产品护照

digital record of product characteristics throughout its life cycle

> 产品全生命周期特征的数字化记录。示例特征包括环境可持续性、环境影响、可回收性。

3.3 unique product identifier 唯一产品标识符

unique string of characters for the identification of a product that also enables a web link to the DPP

> 用于识别产品、并可启用指向 DPP 网络链接的唯一字符串。

3.4 identity proofing 身份核验

process by which the Registration Authority (RA) captures and verifies sufficient information to identify an entity to a specified level of assurance [ISO/IEC 29115:2013, 3.15]

> 注册机构（RA）采集并核验充分信息以按指定保证水平识别实体的过程。

3.5 level of assurance 保证水平

description of the strength of entity authentication assurance [ISO/IEC 24760-1:2025, 3.4.11]

> 实体认证保证强度的描述（ISO/IEC 29115 规定了多级保证水平）。

3.6 digital certificate / certificate 数字证书

public key information of an entity signed by the certification authority (CA) and thereby rendered unforgeable [ISO/IEC 9798-1]

> 由认证机构（CA）签署、将实体公钥信息绑定其中从而不可伪造的证书。

3.7 data carrier 数据载体

device or medium used to store data as a relay mechanism in an automatic identification and data capture system [EN 18220:2026]

> 在自动识别与数据采集（AIDC）系统中作为中继机制用于存储数据的设备或介质。

3 术语和定义（续）/ Terms and definitions (cont.)

3.8 signature / digital signature 签名 / 数字签名

data appended to, or a cryptographic transformation of, a data unit that allows the recipient to prove the origin and integrity of the data unit and protect against forgery. May be used for authentication, data integrity, and non-repudiation. [ISO/IEC 19784-1:2018, 4.34]

> 附加于数据单元或对其作密码变换的数据，使接收方能证明数据单元的源头与完整性并防伪造。可用于认证、数据完整性、不可否认性。

3.9 unique economic operator identifier 唯一经济运营商标识符

unique string of characters for the identification of an actor involved in a product's value chain

> 识别参与产品价值链主体的唯一字符串。

3.10 additional software 附加软件

additional application, program, or tool that a user will install or access separately from the default setup of most smartphones or similar devices to interact with the DPP. Standard functionalities such as web browsers, camera-based QR readers, and universal protocols (DNS, HTTP, HTTPS) are not additional software.

> 用户为与 DPP 交互，需另行安装或访问、不同于大多数智能手机默认配置的应用/程序/工具。网页浏览器、相机二维码读取器、通用协议（DNS/HTTP/HTTPS）不视为附加软件。

3.11 non-repudiation 不可否认性

ability to prove an action or event has taken place, so that this event or action cannot be repudiated later [ISO/IEC 27000]

> 能够证明某行为或事件已发生，使该事件或行为事后不能否认的能力。

3.12 personally identifiable information (PII) 个人可识别信息

information that can be used in a given context to identify, contact, or locate a single person, or to identify an individual in context [ISO 19414:2020, 3.1]

> 在给定语境下可用于识别、联系、定位单个自然人，或在语境中识别个体的信息。

3.13 verifier 验证方

entity that performs the verification process [ISO/IEC 23264-1:2021, 3.27]

> 执行验证过程的实体。

3.14 economic operator 经济运营商

manufacturer, authorized representative, importer, distributor, dealer or fulfilment service provider that places the product onto the market

> 将产品投放市场的制造商、授权代表、进口商、分销商、经销商或履约服务商。

3.15 controlled DPP data 受控 DPP 数据

information on DPP whose access is controlled based on the user's access rights. (User: person who interacts with a system, product or service.) [ISO 26800:2011, 2.10, modified]

> 基于用户访问权限控制其访问的 DPP 信息（用户：与系统、产品或服务交互的人）。

3.16 DPP service DPP 服务

means by which DPP data is made accessible or presented

> 使 DPP 数据可访问或被呈现的方式/服务。

4 DPP 通用安全假设 / General security assumptions for DPP

4.1 通则 / General

The DPP provides required information for products. Product information consists of public and controlled data. All DPP data shall be protected against unauthorised modification and shall be attributable to a verifiable source. The mechanisms ensuring integrity and authenticity of DPP data shall be independent from the security mechanisms used for the communication channels over which the data are transmitted.

> DPP 提供产品的所需信息。产品信息由公开数据与受控数据组成。所有 DPP 数据应防止未授权修改，并可归属于可验证的来源。保障 DPP 数据完整性与真实性的机制，应独立于数据传输所用通信通道的安全机制。

NOTE 1：唯一产品标识的存在是数据可验证性的前提。NOTE 2：ISO 22373/22380/22383 概述应对实物产品欺诈的原则（真实性、完整性、信任），涵盖从假冒到篡改、替换的各类风险。

4.2 系统参与方 / Actors of the system

Actors within the DPP system are defined in EN 18239:2026. The unique economic operator identifier is defined in 3.9 and EN 18219:2026. All actors accessing controlled DPP data shall be identified by a globally unique identifier. Authentication processes shall be based on the required level of assurance.

> DPP 系统参与方由 EN 18239:2026 定义。唯一经济运营商标识见 3.9 与 EN 18219:2026。所有访问受控 DPP 数据的参与方应以全局唯一标识识别。认证过程应基于所需保证水平。

4.3 公开 DPP 数据的访问 / Access to public DPP data

Access limited to reading differs from other use cases: No Authentication Required; Privacy Considerations (measures to avoid profiling); Ease of Use and Scalability; No technological barriers (accessible without additional software); Optional Additional Security shall not prevent access without additional software.

> 仅读取目的的访问不同于其他用例：无需认证；需采取措施避免用户画像；易用与可扩展；无技术壁垒（无需附加软件即可访问）；可选的附加安全措施不得阻碍在无附加软件情况下的访问。

4 DPP 通用安全假设（续）

4.4 通信 / Communication

Communication refers to transmission of product-related information between stakeholders, ensuring all parties have access to accurate, relevant and up-to-date data. The communication shall be secure, using encryption and authentication methods per relevant standards to prevent tampering, unauthorised access, or breaches, maintaining integrity and confidentiality. For public data, no encryption or authentication is required. Requirements from ETSI EN 319 401:2024 (for Trust Service Providers) should be applied to DPP service providers. Responsible parties shall establish seamless and interoperable channels per EN 18216:2026.

> 通信指参与方之间传输产品相关信息，确保各方获得准确、相关、最新的数据。通信应安全，采用符合相关标准的加密与认证方法，防止篡改、未授权访问或泄露，维护完整性与保密性。公开数据无需加密或认证。DPP 服务提供商宜应用 ETSI EN 319 401:2024（面向信任服务提供商）的要求。责任方应依 EN 18216:2026 建立无缝、可互操作的通道。

4.5 存于数据载体中的产品标识 / Product identifier as stored in data carrier

The product identifier within the data carrier should be, where relevant, protected by security measures to support authenticity and integrity. Any solution shall be fully interoperable with unique identifiers per EN 18219:2026. At least one data carrier on each product shall allow users to access public DPP data without additional software. DPP data may indicate that the unique identifier is protected and may provide explicit authentication instructions.

> 在相关且适当的情况下，数据载体内产品标识应受安全措施保护以支撑真实性与完整性。任何方案应与 EN 18219:2026 唯一标识完全互操作。每件产品至少一个数据载体应允许用户无需附加软件访问公开 DPP 数据。DPP 数据可标明唯一标识受保护，并可提供明确的核验指引。

4.6 数据载体真实性 / Data carrier authenticity

The economic operator may secure the data carrier; the DPP can include information on how to inspect related security features. Any solution for physical authenticity shall be fully interoperable with data carriers per EN 18220:2026. Such solutions may require additional data carriers and/or data sources. The protected data carrier shall allow all users to access public DPP data without additional software. DPP data may indicate the carrier(s) are protected and provide explicit authentication instructions.

> 经济运营商可加固数据载体；DPP 可包含如何查验相关安全特征的信息。确保实物载体真实性的方案应与 EN 18220:2026 数据载体完全互操作。此类方案可能需要附加载体和/或数据源。受保护载体应允许所有用户无需附加软件访问公开 DPP 数据。DPP 数据可标明载体受保护并提供明确核验指引。

4.7 DPP 真实性与数据完整性 / DPP authenticity and data integrity

A DPP shall be verifiable in terms of authenticity and integrity. Such verification shall be free of charge and based on interoperable mechanisms in regulated scenarios (e.g. cross-border).

> DPP 应在真实性与完整性方面可验证。该验证应免费，并在受监管场景（如跨境）中基于可互操作机制。

All actors providing DPP data shall be authenticated using a mechanism requiring proof of ownership of a digital certificate issued through an identity proofing process compatible with relevant legal acts.

> 所有提供 DPP 数据的参与方，应通过要求证明其持有经与相关法律行为兼容的身份核验流程所签发数字证书的认证机制进行认证。

Every modification (including first creation and deletion) shall be bound to the identity of the authenticated actor (e.g. valid digital certificates), providing non-repudiation. Examples: digital signatures, digital seals, electronic attestation of attributes, ESDCs.

> 每次修改（含首次创建与删除）应绑定执行该修改的已认证参与方身份（如有效数字证书），以提供对该修改的不可否认性。技术手段示例：数字签名、电子印章、属性电子证明、ESDC。

Checking validity and integrity of DPP data shall be free of charge and without limitations for the verifier. A user is not obliged to verify the DPP data.

> 检查 DPP 数据的有效性与完整性应对验证方免费且无限制。用户无义务验证 DPP 数据。

Any creation, modification, or deletion shall be logged in a non-repudiable, tamper-proof manner; log data integrity shall be provided over time without additional supporting documents or offline procedures. The responsible editor shall be identifiable via a globally unique operator identifier. Any operational-phase event modifying a DPP shall be logged in an audit-proof manner, accessible by relevant parties.

> 任何创建、修改或删除应以不可否认、防篡改的方式记入日志；日志数据完整性应长期维护，无需附加支撑文件或离线流程。责任编辑者应可通过全局唯一运营商标识识别。运行期任何修改 DPP 的事件应以可审计方式记录，供相关方访问。

5 唯一标识、载体与 DPP 数据的风险与要求

5.1 数据安全与隐私风险 / Risk on data security and privacy

5.1.1 通则 / General

These subclauses refer to anonymous access to public data using common devices (e.g. smartphones with built-in capabilities) without additional software, and to authenticated access to controlled data by authorised user groups, including CRUD operations.

> 本分条款涉及使用常用设备（如带内置能力的智能手机）无需附加软件匿名访问公开数据，以及授权用户组对受控数据的认证访问，含 CRUD（增删改查）操作。

5.1.2 个人数据保护 / Protection of personal data

Implementation shall not gather any personal information from natural persons accessing public data only. In case of access to restricted data, authentication information may be requested.

> 仅访问公开数据的实现不得收集自然人的任何个人信息。访问受限数据时，可要求提供认证信息。

5.1.3 个人数据安全 / Security of personal data

Where personal data are gathered for permitted reasons, it shall be secured against data loss, disallowed access and any use not covered by permitted reasons. Implementation/operator shall provide legally required information, correction and deletion. DPP service shall not use external analytic tools and shall not include any reference to external components in the response.

> 为获准理由收集个人数据时，应防止数据丢失、未授权访问及超出获准理由的使用。实现方/运营商应提供法律要求的告知、更正与删除途径。DPP 服务不得使用外部分析工具，响应中不得含任何外部组件引用。

5.1.4 组织数据安全 / Security of organizational data

Robust measures (encryption, access controls, regular audits) should be implemented. Organisational data shall be protected against unauthorised access, disclosure, alteration and loss; appropriate measures shall ensure confidentiality, integrity, availability. Security measures shall be consistent with recognised standards, and shall secure public-data access without unnecessary barriers to interoperability, fair competition, or legitimate-actor access.

> 应实施稳健措施（加密、访问控制、定期审计）。组织数据应防未授权访问、披露、篡改与丢失；应采取措施保障保密性、完整性、可用性。安全措施应符合公认标准，并在不造成互操作、公平竞争或合法主体访问不必要壁垒的前提下保障公开数据访问。

5.1.5 防止画像 / Prevention of profiling

Combining IP addresses, HTTP protocol data, and user behaviour with DPP access may facilitate profiling in violation of privacy; such practices shall be proactively prevented where technically feasible. User profiling shall not be allowed unless authentication is strictly required to access a specific service.

> 将 IP 地址、HTTP 协议数据与用户行为结合 DPP 访问可能助长违反隐私的用户画像；在技术可行处应主动防止此类做法。除非访问特定服务严格需要认证，否则不允许用户画像。

5.1.6 确保安全用户访问 / Ensuring safe user access

When providing access to public data, the DPP system shall never request personal data or PII from the user's device. The system shall not access any advanced functionalities (including but not limited to position, notifications, context/network information, call/messaging management, application installation, network/device configuration changes).

> 提供公开数据访问时，DPP 系统不得索取用户设备的个人数据或任何 PII。系统不得访问任何高级功能（包括但不限于定位、通知、上下文/网络信息、通话/短信管理、应用安装、网络/设备配置更改）。

注：若让用户知晓访问公开数据绝不需要输入数据或授予执行权限，可提升安全性，并可提供补充说明。

5.1.7 防钓鱼、quishing 与恶意代码 / Protection against phishing, quishing, and malicious code

Phishing deceives users into disclosing sensitive information; quishing exploits QR codes etc. to redirect to malicious sites. Protection may also address unauthorised copying of data carriers containing unique identifiers (likely in counterfeit or non-conforming products). Mitigations: promote user awareness and safe usage, rely on trusted and verifiable entry points, use anti-malware tools, network monitoring, secure protocols.

> 钓鱼诱骗用户泄露敏感信息；quishing 利用二维码等将用户导向恶意站点。防护还可针对含唯一标识的数据载体被未授权复制（多见于假冒或不合规产品）。缓解措施：提升用户意识与安全使用、依赖可信可验证入口、使用反恶意软件、网络监控、安全通信协议。

5.1.8 防批量抓取 / Safeguarding against mass data scraping

Mass scraping risks integrity and security by extracting sensitive product/consumer information at scale. Robust anti-scraping mechanisms should be implemented. The system may allow beneficial search engines to index DPP data. Examples: rate limiting, access controls. Continuous monitoring and adaptive measures are necessary.

> 批量抓取可大规模提取敏感产品/消费者信息，危及系统完整性与安全。应实施稳健反抓取机制。系统可允许有益的搜索引擎索引 DPP 数据。示例：限流、访问控制。需持续监控与自适应安全措施。

5.2 标识风险与要求 / Risk and requirements on identification

5.2.1 通则 / General

Accurate identification of products, manufacturers and other actors is crucial to ensure integrity. Risks arise from incorrect/fraudulent identification, leading to circulation of unauthorised or counterfeited goods and misrepresentations in the value chain, affecting trust and safety. Appropriate reliable, secure and verifiable identification mechanisms may be implemented, leveraging unique identifiers and secure data protocols. Identifiers and data carriers shall follow EN 18219:2026 and EN 18220:2026.

> 准确识别产品、制造商及其他参与方对保障完整性至关重要。错误/欺诈性标识会导致未授权或假冒产品流通及价值链中的虚假陈述，影响信任与安全。可采用适当、可靠、安全、可验证的标识机制，利用唯一标识与安全数据协议。标识与数据载体应遵循 EN 18219:2026 与 EN 18220:2026。

5.2.2 责任主体的认证与追溯 / Authentication and traceability of responsible actors

Any actor granted access to controlled data beyond read-only public access shall be identifiable per EN 18239:2026. Authentication and identification shall not be restricted based on the location of natural or legal persons, as far as legal requirements are fulfilled.

> 任何获得超出公开只读数据访问权限的受控数据访问权的参与方，应按 EN 18239:2026 可识别。在满足法律要求的前提下，认证与标识不得基于自然人/法人的所在地加以限制。

5.3 产品风险 / Risk to products

5.3.1 防假冒 / Protecting against counterfeiting

Where relevant, a counterfeiting risk assessment should be conducted. Where elevated risk is identified, proportionate mitigation may be implemented, including protecting the data carrier with authentication elements verifiable by stakeholders (including consumers) e.g. via visible indicators and/or a DPP-linked authentication service.

> 在相关处，宜进行假冒风险评估。识别出较高风险时，可采取相称的缓解措施，包括用可由相关方（含消费者）核验的认证要素加固数据载体，例如通过可见指示和/或 DPP 关联认证服务。

5.3.2 DPP 数据完整性风险 / Risks related to DPP data integrity

Retrospective tampering shall be prevented by recording modifications to regulatory-relevant content per EN 18221:2026. Misuse of a valid DPP with counterfeit products may lead to health, environmental and economic risks. Where justified by value-cost relation, relevant actors should include functionality to detect suspicious requests and initiate alerts/investigation.

> 应按 EN 18221:2026 记录对监管相关内容所做的修改，防止事后篡改。在假冒产品上误用有效 DPP 可能带来健康、环境与经济的重大风险。在成本收益合理且适用处，相关参与方可纳入检测可疑请求并触发告警/调查的功能。

5.4 防排斥的风险与要求 / Risk and requirements to prevent exclusion

5.4.1 通则 / General

The system shall be accessible to all users, regardless of technological capabilities, disabilities, or socioeconomic status. Stakeholders should be trained to understand and accommodate diverse user needs to minimise exclusion risk.

> 系统应让所有用户可访问，不论其技术能力、残障状况或社会经济地位。应培训参与方理解并满足用户的多样化需求，以最小化排斥风险。

5.4.2 任意消费设备访问 DPP / Accessibility to a DPP with any consumer device

Every natural person shall be able to access a DPP at any time using an internet-connected device (smartphone, tablet, computer). Essential functionalities shall be easily accessible through commonly available web browsers, without requiring installation or use of additional software.

> 每个自然人应能随时使用联网数字设备（智能手机、平板、电脑）访问 DPP。核心功能应可通过常用网页浏览器轻松访问，无需安装或使用附加软件。

5.4.3 残障人士可访问性 / Accessibility for persons with disabilities

DPP implementations shall enable persons with disabilities to perceive, understand, navigate and interact with public DPP content. Accessibility may be supported through compatibility with assistive technologies and adaptable presentation. The data carrier may include features to facilitate location and scanning. Accessibility measures shall not weaken security mechanisms; alternative access modalities shall preserve data integrity and authenticity; security mechanisms should not prevent persons with disabilities from accessing DPP information.

> DPP 实现应使残障人士能够感知、理解、导航并交互公开 DPP 内容。可通过兼容辅助技术与可适配呈现来支撑可访问性。数据载体可包含便于定位与扫描的特征。可访问性措施不得削弱安全机制；替代访问方式应保留数据完整性与真实性；安全机制不应阻止残障人士访问 DPP 信息。

> **第 5 章要点速览：** 公开数据匿名访问、不收集个人信息、不索取 PII、防画像/防钓鱼/防抓取；标识与载体须符合 EN 18219/18220；受控数据访问者须可识别；防假冒风险评估；残障人士与任意设备可访问；安全措施不成为排斥因素。

Annex A（normative 规范性）

电子签名数据构造（ESDC）/ Electronically signed data constructs (ESDCs)

A.1 通则 / General

This annex provides the essential aspects of ESDCs that enable secure and verifiable digital representations of product data and credentials.

> 本附录给出 ESDC 的基本内容，用于实现产品数据与凭证的安全、可验证数字化表示。

A.2 引言 / Introduction

ESDCs are electronically signed documents or data sets, used to create a portable and verifiable digital representation of product-related information, attestations, and credentials. They are cryptographically signed by their issuing party, making them tamper-evident and digitally verifiable. ESDCs are a primary means for establishing trust in the DPP ecosystem.

> ESDC 是电子签名的文档或数据集，用于创建产品相关信息、证明与凭证的可携带、可验证数字化表示。由签发方作密码签名，使其具备篡改可见性与数字可验证性。ESDC 是在 DPP 生态中建立信任的主要手段。

A.3 关键特征 / Key characteristics

ESDCs provide a mechanism to:

> ESDC 提供以下机制：

- **verify DPP data integrity**：确保签发后数据未被改动
- **verify DPP data authenticity**：确认签发方身份及数据来源
- **provide non-repudiation of issuer**：提供签发方的不可否认性
- **verify issuer authority and accreditation**：通过信任清单/治理框架核验签发方证书状态与属性

An ESDC generally involves: **Issuer** (entity asserting claims, identified via recognised identifier linked to signing key/certificate; typically the economic operator placing the product on the market); **Subject** (the product, identified by a globally unique product identifier); **Data/Attributes** (e.g. material composition, recycling advice, conformity certificates); **Signature** (electronic signature ensuring integrity and authenticity, generated with issuer's private key, certificate managed within a recognised trust framework).

> ESDC 一般包含：**签发方 Issuer**（主张声明/数据的实体，通过关联签名密钥/证书的可识别标识识别；核心产品数据通常由投放市场的经济运营商签发）；**主体 Subject**（产品，由全局唯一产品标识识别）；**数据/属性 Data/Attributes**（如材料组成、回收建议、合格证书）；**签名 Signature**（保障完整性与真实性的电子签名，由签发方私钥生成，证书在公认信任框架内管理）。

ESDCs can be implemented using various technologies; choice depends on use case, data carrier constraints, existing infrastructure, and regulatory requirements. Examples: ISO 22376, ISO/IEC 20248, W3C Verifiable Credentials, Electronic Attestation of Attributes, or electronic seals and signatures.

> ESDC 可用多种技术实现，选择取决于用例、载体约束、既有基础设施与监管要求。示例：ISO 22376、ISO/IEC 20248、W3C 可验证凭证、属性电子证明、电子印章与签名。

Annex B（informative 资料性）

ESDC 示例 / Examples of electronically signed data constructs

B.1 通则 / General

This annex provides a non-exhaustive set of informative descriptions of specific approaches and standards that can be used to implement ESDCs. Choice depends on use case, regulatory mandates, existing infrastructure and data carrier constraints.

> 本附录非穷举地给出可用于实现 ESDC 的具体方法与标准的说明性描述。选择取决于用例、监管规定、既有基础设施与数据载体约束。

B.2 属性电子证明 / Electronic attestation of attributes (EAA)

Introduction: EAA is a trust service providing a legally recognised mechanism for digitally attesting and verifying attributes of natural persons, legal persons, or objects.

> **引言：**属性电子证明（EAA）是一种信任服务，为自然人、法人或对象的属性提供法律认可的数字证明与验证机制。

Key characteristics: Issuer (in DPP context, a level of assurance and legal recognition of issuer is expected; typically the economic operator); Subject (product, globally unique identifier per EN 18219); Attributes/Claims (e.g. material composition, conformity status, repair history); Format and Signing (digitally signed; W3C VCs widely considered suitable); Verification (via issuer's public key/certificate and trust framework); Legal Value (derived from issuer credentials, signed-data integrity, validity info).

> **关键特征：**签发方（DPP 语境下应有保证水平与法律认可，通常为经济运营商）；主体（产品，EN 18219 全局唯一标识）；属性/声明（如材料组成、合格状态、维修历史）；格式与签名（数字签名，W3C VC 普遍被视为合适格式）；验证（用签发方公钥/证书及信任框架）；法律价值（来自签发方凭证、签名数据完整性与有效性信息）。

B.3 可见数字印章 VDS（ISO 22376）

Introduction: VDS, compliant with ISO 22376, is a representation of electronically signed data optimised for visual carriers (e.g. 2D barcodes/QR codes), designed tamper-evident and digitally verifiable. Trust is achieved via an electronic signature anchored in a governance framework (e.g. ISO 22385).

> **引言：**VDS 符合 ISO 22376，是电子签名数据的表示形式，为可视载体（如二维码）优化，具备篡改可见与数字可验证性。信任通过锚定在治理框架（如 ISO 22385）中的电子签名实现。

Risk mitigation: authenticity of the link to DPP; integrity of embedded product identifier; reliable physical-to-digital link; legitimacy of declarations. **Structure:** Seal issuer; Payload (header + essential data such as unique product identifier, secure web link, critical indicators); Electronic Signature. Can often be verified offline with pre-loaded trust information.

> **降险：**DPP 链接真实性；嵌入产品标识完整性；物理—数字可靠绑定；声明合法性。**结构：**印章签发方；载荷（头 + 唯一产品标识、安全网页链接、关键指标等核心数据）；电子签名。凭预加载信任信息常可离线验证。

B.4 AIDC 数字签名可验证数据结构（DigSig — ISO/IEC 20248）

Introduction: ISO/IEC 20248 specifies a Digital Signature Data Structure (DigSig) for AIDC techniques, to digitally sign, structure, encode, and verify data within carriers like 1D/2D barcodes, RFID and NFC tags, supporting offline verification where PKI elements are available.

> **引言：**ISO/IEC 20248 规定用于 AIDC 技术的数字签名数据结构（DigSig），对条码（1D/2D）、RFID、NFC 等载体中的数据作数字签名、结构化、编码与验证，在具备 PKI 要素时支持离线验证。

Key characteristics: Issuer/Domain Authority (DA, identified by DAID often based on ISO/IEC 15459); Data structure (hierarchical JSON-based DigSig Data Description DDD); Encoding and snips (signed data encoded into segments distributable across carriers); Certificates (X.509 as trust anchors); Offline verification; Interoperability framework (track-and-trace, IoT authentication, document verification).

> **关键特征：**签发方/域权威 DA（由 DAID 识别，常基于 ISO/IEC 15459）；数据结构（层级化 JSON DigSig Data Description）；编码与 snip（签名数据编码为可在载体间分发的分段）；证书（X.509 作信任锚）；离线验证；互操作框架（追踪追溯、IoT 认证、文件验证）。

DigSig as ESDC for DPP: encode essential DPP data in DDD; secure with DA's digital signature; embed signed data onto AIDC carriers; verifiers parse DigSig and verify signature against DA certificate; support lifecycle management/update of signed data if carrier supports updates.

> **DigSig 作为 DPP 的 ESDC：**在 DDD 中编码核心 DPP 数据；用 DA 数字签名保护；将签名数据嵌入产品 AIDC 载体；验证方解析 DigSig 并用 DA 证书验证签名；若载体支持更新，可管理产品全生命周期的签名数据。

B.5 W3C 可验证凭证 / W3C Verifiable Credentials

Introduction: A tamper-proof digital statement issued by an authority about a subject, cryptographically verifiable by any party.

> **引言：**由权威方就某主体签发的防篡改数字声明，任何参与方均可密码学验证。

Key characteristics: Authenticity; Integrity; Verifiability (any verifier independently, without contacting issuer); Interoperability (open standards); Privacy and selective disclosure (e.g. zero-knowledge proofs).

> **关键特征：**真实性；完整性；可验证性（任何验证方独立验证，无需联系签发方）；互操作性（开放标准）；隐私与选择性披露（如零知识证明）。

Implementation as ESDC: Issuer (economic operator issues VC with product data); Holder (DPP system stores VC); Verifier (application, customer, customs, third party verifies locally and free of charge). VCs may be embedded or referenced via secure links, and can be chained/aggregated for complex histories.

> **作为 ESDC 的实现：**签发方（经济运营商签发含产品数据的 VC）；持有方（DPP 系统代为存储）；验证方（应用、客户、海关、第三方在本地免费验证）。VC 可嵌入或通过安全链接引用，并可串联/聚合以表达多主体复杂历史。

B.6 数字签名与电子印章 / Digital signatures and electronic seals

Introduction: A digital signature provides signer authentication, data integrity (tamper-evident), and non-repudiation. Based on standards: ETSI EN 319 122-1 (CAdES), 319 132-1 (XAdES), 319 142-2 (PAdES), TS 119 182-1 (JAdES). Requires signer possession of a private key whose public key is in a certificate enrolled by a recognised CA.

> **引言：**数字签名提供签名者认证、数据完整性（篡改可见）与不可否认性。基于标准：ETSI EN 319 122-1（CAdES）、319 132-1（XAdES）、319 142-2（PAdES）、TS 119 182-1（JAdES）。要求签名者持有私钥，其对应公钥在公认 CA 签发的证书中。

Main difference: digital signatures provide electronic signatures for natural persons; electronic seals provide electronic signatures for legal persons.

> **主要区别：**数字签名面向自然人提供电子签名；电子印章面向法人提供电子签名。

Digital certificates: bind a natural/legal person's identity to a cryptographic key. Certificate may be stored locally or accessed remotely. Trust Service Providers within trust frameworks define roles, responsibilities, and technical requirements, enabling interoperability and mutual recognition between trust domains.

> **数字证书：**将自然人/法人的身份绑定到密码学密钥，可本地存储或远程访问。信任框架内的信任服务提供商定义角色、责任与技术要求，实现信任域间的互操作与互认。

Annex C（informative 资料性）

公平竞争相关风险与考量 / Risk and considerations related to fair competition

C.1 General: DPP implementation shall ensure all market participants have equal access to the resources, tools, and data required for compliance.

> **C.1 通则：**DPP 实施应确保所有市场参与者平等获取合规所需的资源、工具与数据。

C.2 Avoiding market restrictions: Requiring vendor-specific or commercial software could lead to market restrictions. The basic functionality of searching and reading public DPP data shall be implementable without installing commercial/vendor-specific software.

> **C.2 避免厂商软件造成市场限制：**要求厂商专有或商业软件可能造成市场限制。检索与读取公开 DPP 数据的基本功能应能在不安装商业/厂商专有软件的情况下实现。

C.3 Disproportionate resource demand for SMEs: core functionalities shall be provided in an easily accessible way; voluntary additional functionalities shall complement, not replace, core functionalities.

> **C.3 避免给中小企业造成不成比例的资源负担：**核心功能应以易获取方式提供；自愿附加功能应补充而非替代核心功能。

Annex ZA（informative 资料性）

与法规 (EU) 2024/1781 的关系 / Relationship with Regulation (EU) 2024/1781

This standard has been prepared under the Commission's standardisation request M/604/C(2024)5423 to provide one voluntary means of conforming to Regulation (EU) 2024/1781 (Ecodesign for Sustainable Products Regulation, ESPR). Once cited in the Official Journal of the EU, compliance with the normative clauses in Table ZA.1 confers, within the scope of this standard, a presumption of conformity with the Regulation and associated EFTA regulations.

> 本文件依欧委会标准化请求 M/604/C(2024)5423 编制，作为符合《欧盟 2024/1781 号法规》（可持续产品生态设计法规 ESPR）的一种自愿手段。一旦在欧盟官方公报（OJEU）被引用，在本标准范围内符合表 ZA.1 规范性条款即推定符合该法规及相关 EFTA 法规。

Table ZA.1 — 本文件与 (EU) 2024/1781 第 10、11 条的对应关系

EU 2024/1781 要求	本 EN 条款	备注
Article 11(a)	4.2	限于端到端通信与数据传输的组织互操作

WARNING 1: Presumption of conformity stays valid only as long as the reference is maintained in the OJEU list. WARNING 2: Other Union legislation may be applicable.

> 警告 1：仅当本标准的引用保留在 OJEU 列表中时，合规推定才有效。警告 2：其他欧盟立法可能同样适用。

参考文献 / Bibliography（节译）

#	标准/文献	中文名
[1]	ISO/IEC 29115:2013	信息技术—安全技术—实体认证保证框架
[2]–[7]	ISO 22380 / 22383 / 22385 / 22373 / 22376 / 22387	安全与韧性—产品与文件真实性、完整性、信任（风险原则/认证方案选择/信任互操作框架/供应链/ VDS /实物度量验证）
[8]	ISO/IEC 20248	AIDC 数字签名数据结构模式
[9]–[13]	ETSI EN 319 122-1 / 132-1 / 142-2 / TS 119 182-1 / EN 319 401:2024	CAdES / XAdES / PAdES / JAdES 签名；信任服务提供商通用政策要求
[14]–[15]	W3C VC Data Model 1.1 / 2.0	W3C 可验证凭证数据模型
[16]	ISO/IEC 24760-1:2025	信息安全管理—身份管理框架—第1部分：核心概念与术语
[17]–[18]	ISO/IEC 9798-1 / 19784-1:2018	实体认证—第1部分；生物识别 API

#	标准/文献	中文名
[19]–[23]	ISO/IEC 27000 / ISO 19414:2020 / 23264-1:2021 / 26800:2011 / 15459	信息安全管理；智能交通；数据脱敏；人因工学；AIDC